

AATHII RAKURAKAVAN

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University of Toronto Honours B.Sc. graduate (Computer Science & Mathematics minors) building IRL, a full-stack iOS fitness app with a Python/FastAPI backend, Gemini-powered computer vision, and App Store subscription infrastructure. Systems background in C, Haskell, and Racket; two years of client-facing sales and operations work. Trilingual in English, French, and Tamil.

EDUCATION

University of Toronto — Toronto, ON

Sept 2021 – Apr 2026

Honours Bachelor of Science, Technology, Coding & Society | Double Minor: Computer Science and Mathematics

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, Java, SQL, Haskell, Racket, Bash

Frameworks & Libraries: React Native, React, Expo, FastAPI, Pydantic, React Navigation

AI & Computer Vision: Google Gemini API, OpenAI SDK, OpenCV (YuNet, ORB), image-generation pipelines

Backend & Data: REST API design, SQLite, JWT / OAuth authentication, third-party API integration

Infrastructure: Render, Linux/Unix, Expo Application Services (EAS), Excel automation

Developer Tools: Git, Claude Code, OpenAI Codex, pytest, ESLint, GDB, AddressSanitizer

PROJECTS

IRL — AI Fitness & Nutrition iOS App | React Native, Expo, Python, FastAPI, SQLite

May 2026 – Present

- Architected a full-stack iOS application: an Expo/React Native client (Expo SDK 54, React 19, React Navigation 7) against a Python/FastAPI REST backend on Render with persistent SQLite storage; automated builds and TestFlight distribution via Expo Application Services.
- Integrated Google Gemini for AI meal scanning, personalized coaching, and image generation, and built a local OpenCV/YuNet + ORB face-verification step so progress-photo identity checks run on-device instead of via a third-party service.
- Implemented authentication with Apple and Google Sign-In, JWT (HS256) session tokens, PBKDF2 password hashing, SecureStore token storage, and Resend-backed password-reset email flows.
- Shipped native iOS integrations — Apple HealthKit, Camera, Photo Library, and APNs push notifications — plus subscription monetization managed through RevenueCat.
- Unified nutrition data from USDA FoodData Central, Open Food Facts, and FatSecret behind one lookup service, added ElevenLabs voice coaching over a Twilio number, and covered the backend with a 119-test pytest suite.

Linux System Shell | C, Sockets, GDB

Apr 2025

- Built a functional Unix shell (mysh) in C implementing core utilities (ls, cd, cat, wc) and managing process lifecycles with fork/exec for background execution and signal handling.
- Added client-server communication over TCP/UDP sockets and verified zero memory leaks across manual allocation paths using AddressSanitizer and GDB.

Functional Language Interpreter & Desugarer | Haskell, Racket

Apr 2025

- Implemented an interpreter for Orange, a functional language, with an abstract syntax tree, eager evaluation, environment-based variable lookup, closures, and recursive evaluation with structured error propagation.
- Wrote a structural-recursion desugarer in Racket translating high-level "Sugar" syntax into a core language, handling n-ary function calls and deeply nested pattern matching.

PROFESSIONAL EXPERIENCE

Sales Lead — Fitness Connection, Richmond Hill, ON

May 2025 – Present

Promoted from Sales Specialist

- Built an automated workflow against the GleanTap CRM API that auto-logs texts, emails, and notes and self-completes routine follow-up tasks, saving several hours of manual data entry per week and reducing the daily workflow to phone calls alone.
- Ranked top-performing sales representative at the location; own the full membership sales cycle from first inquiry through close.
- Work across three CRM platforms — GleanTap, Antaris, and Less Annoying CRM — reconciling lead, appointment, and conversion records between systems and resolving member account and billing issues.

Event Support Lead — DUA Events, University of Toronto, Toronto, ON

May 2023 – Jun 2025

- Coordinated logistics, setup, teardown, and live issue resolution across teams for campus events on tight timelines, troubleshooting A/V and operational failures and communicating shifting priorities to staff, vendors, and attendees.